Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/07/2005
PI Name:	Acrivos, Andreas Ph.D.		
Project Title:	Particle Segregation in a Flowing Concentrated Suspension Subject to High-Gradient Strong Electric Fields		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10031	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Khusid, Boris (New Jersey Institute of Technology)		
Grant/Contract No.:	NAG3-2698		
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Task Description:	Aside from the scientific challenges, the widespread use of electro-hydrodynamic devices and processes emphasizes a critical need for developing a comprehensive predictive theory capable of improving our fundamental understanding of the behavior of a suspension subject to an electric field and shear, and of facilitating the design and optimization of such devices. The general feature of all these phenomena is that the application of a sufficiently strong field induces the motion of particles and fluid caused by a transfer of energy and momentum from the electric source. The redistribution of the constituents driven by the suspension flow affects, in turn, the applied field by changing the system polarization and thus rendering the electric and hydrodynamic phenomena strongly coupled. Our study shall constitute a theoretical framework for electro-hydrodynamics of suspensions. The focus will be on physical processes - field- and flow-induced transitions, suspension flow, particle motions, electrical forces, interparticle electric-and hydrodynamic interactions, etc The purpose of the research is: 1) To test/develop/generalize the currently available theories of constitutive relations for the electric energy and stress in dispersive and dissipative multi-component systems. 2) To develop a theory for the dynamics of field induced phase transitions in concentrated flowing suspensions, including the collective phenomena of particle motions and segregation, shear-induced diffusion, diffusion-convection, and phase separation. 3) To develop a theory of the electro-hydrodynamics of suspensions which includes the coupled effects of the electrorheological response, dielectrophoresis, and shear-induced diffusion and re-suspension of concentrated suspensions under normal and low-gravity conditions, applicable to a broad range of particle polarizability which involves high particle/fluid density ratios. 4) To perform rigorous experimental tests of the theoretical predictions in ground-based experiments involving slow-rotating electric chambers in which the time averaged gravity force acting on a flowing suspension is zero in the rotating framework. 5) To perform rigorous tests of the theoretical predictions in KC-135 and in long-duration micro-gravity space experiments having true instantaneous near-zero gravity forces in order to investigate the field-driven phase-transition in a gravity-free environment and better estimate the effects of gravity-induced flows in ground-based experiments. 6) To generate new experimental data on the field-induced particle motions, segregation, and phase transitions in a flowing suspension with application to electro-hydrodynamic micro-fluidic devices, actuators, separators, and bio-reactors. The main outcome of the proposed theoretical and experimental studies shall be a theoretical framework for strongly coupled electric and hydrodynamic phenomena in a concentrated suspension subject to strong electric fields and shear. If successful, this research shall bring the understanding of the fundamentals of suspension electro-hydrodynamics to the level where it will become useful in the design of electro-hydrodynamic apparatuses and processes for a broad range of industrial, biological, and environmental applications. This will stimulate further "spin-offs" of electro-technologies. The proposing team has already been engaged in two engineering applications of electro-hydrodynamics. To-date, based on preliminary experimental and theoretical results which we have obtained, we expect that the proposed research has the strong potential for the elucidation of new electric and hydrodynamic phenomena caused by interparticle interactions in concentrated suspensions that are absent when these effects operate separately. The salient features embodied in our theory is similarity between the electric-field-driven transitions in suspensions and the first-order phase transformations in quenched conventional binary systems. A validation of this similarity for flowing suspensions will accelerate studies of the dynamics of the field-driven processes by using the currently available knowledge concerning the behavior of quenched binary systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	If successful, this research shall bring the understanding of the fundamentals of suspension electro-hydrodynamics to the level where it will become useful in the design of electro-hydrodynamic apparatuses and processes for a broad range of industrial, biological, and environmental applications. This will stimulate further "spin-offs" of electro-technologies.
Task Progress:	1. Experiments were conducted on concentrated suspensions of neutrally buoyant particles which exhibit negative dielectrophoresis. We found that, due to interparticle electrical interactions, such suspensions undergo a phase separation when subjected to a high-gradient ac field (~ kV/mm), and form a propagating distinct front between the regions enriched with and depleted of particles. A generalization of our theory for the thermodynamics of the field-induced phase transitions in suspensions of polarized particles [Phys. Rev. E, 52, 1669 (1995), 54, 5428 (1996), and 60, 3015 (1999)] is proposed for the front propagation, and its predictions are shown to be consistent with the experiments even though the model contains no fitting parameters. The combination of field-induced dielectrophoresis and phase transition provides a new method for strongly concentrating particles in prespecified regions of dielectrophoretic devices. 2. Experiments in microgravity (aboard the NASA research aircraft KC-135) and ground-based environments were conducted in a clinostat (a channel slowly rotated around its horizontal axis) in which a dilute suspension of heavy, positively polarized spheres was exposed to a high-gradient strong ac electric field. The presented data demonstrate that dielectrophoresis provides a powerful method for the manipulation and concentration of particles in clinostats. However, it is found that clinorotation does not simulate the zero gravity morphology of the aggregation pattern due to a pronounced effect of a relatively weak, rotating gravity on the field-driven interparticle interactions. 3. Experiments were conducted in microfluidics equipped with dielectrophoretic gates arranged perpendicular to the flow. Under the action of a high-gradient ac field and shear, flowing suspensions were found to undergo a phase separation and to form a distinct front between the regions enriched with and depleted of particles. We demonstrate that this many-body phenomenon, which originates from interparticle electrical interactions, provides a new method for concentrating particles in focused regions and for separating biological and non-biological materials, a critical step in the development of miniaturized biological assays. The evolution of the particle patterns formation is well described by a proposed electro-hydrodynamic model.
Bibliography Type:	Description: (Last Updated: 02/28/2020)

Articles in Peer-reviewed Journals	Kumar A, Qiu Z, Acrivos A, Khusid B, Jacqmin D. "Combined negative dielectrophoresis and phase separation in nondilute suspensions subject to a high-gradient ac electric field." Physical Review E. 2004 Feb;69(2 Pt 1):021402. PubMed PMID: 14995439 ; https://doi.org/10.1103/PhysRevE.69.021402 , Feb-2004
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Articles in Peer-reviewed Journals	Bennett D, Khusid B, Galambos P, James CD, Okandan M, Jacqmin D, Acrivos A. "Combined field-induced dielectrophoresis and phase separation for manipulating particles in microfluidics." Applied Physics Letters. 2003 Dec;83(23):4866. https://doi.org/10.1063/1.1629789 , Dec-2003
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Presentation	B. Khusid, A. Acrivos "Field- and shear-driven collective phenomena in suspensions" International Electrokinetics Conference , Pittsburgh, PA, June 13-17, 2004 Jun-2004
Presentation	A. Acrivos, A. Kumar, Z. Qiu, B. Khusid, D. Jacqmin, "Front formation in a suspension of negatively polarized particles subjected to a high-gradient ac electric field," AIChE Annual Meeting, November 16-21, 2003, San Francisco, CA Nov-2004
Presentation	D.J. Bennett, B. Khusid, P.C. Galambos, C.D. James, M. Okandan, "An integrated microfluidic dielectrophoretic particle and cell concentrator, " 2003 ASME Mechanical Engineering Congress, November 15 – 21, Washington, DC Nov-2004
Presentation	B. Khusid, A. Acrivos, "Electric-field-driven phenomena for manipulating particles in micro-devices" The Seventh Microgravity Fluid Physics and Transport Phenomena, Cleveland, OH. June 22-23, 2004. Jun-2004
Presentation	B. Khusid, "Simulations of micro- and nano-scale manipulation of biomolecules with high-gradient AC fields, " NSF/ Sandia National Labs Principal Investigators' Meeting, January 27-28, 2004, Albuquerque, NM Jan-2004
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